

# SHAIFALI GARG

Analytical Thinker | Quantitative Researcher | BI Manager

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## PROFESSIONAL SUMMARY

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Analytical thinker with strong mathematical foundations and ten years tackling complex problems across quantitative disciplines, from supply chain optimization to predictive modeling. Cross-disciplinary expertise grounded in business analytics. Proven ability to rapidly learn new domains and apply rigorous analytical frameworks. Strong communicator translating complex quantitative concepts for interdisciplinary teams. Curiosity-driven researcher committed to leveraging analytical expertise to build impactful AI systems.

## CORE COMPETENCIES

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Math Modeling, Statistical Analysis, Problem Solving, Quantitative Research, Facilities Services Management, Predictive Analytics, Machine Learning, Rapid Learning, Hypothesis Testing, Regression Analysis, Time Series, Neural Networks

## WORK EXPERIENCE

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### Manager, Business Intelligence and Reporting

2022 - Present

*Facilities Services, Walmart Inc, Bentonville, USA*

- Created new processes and workflows in Phoenix (Quickbase) that help run the business and finance across all POs, maintaining financial changes in data directly from SAP.
- Lead cross-disciplinary teams, fostering collaborative problem-solving and a continuous learning culture.
- Translate complex quantitative insights into clear, actionable recommendations for executive leadership.
- Design advanced analytical solutions leveraging statistical methods, predictive modeling, and ML algorithms.
- Demonstrate rapid learning agility by mastering new domains, tools, and methodologies.
- Manage complex BI and reporting solutions by researching potential solutions and making recommendations to meet end-user objectives; updating business data; creating visual prototypes and design documentation; monitoring solutions and troubleshooting issues; completing dashboard and reporting specifications; and creating reports and visualizations to effectively communicate results.
- Manage the implementation of BI solution roadmaps by developing written and oral presentations for diverse audiences (management, customers, suppliers, BI and analytics staff); ensuring solution testing and validation; ensuring requirements are met to support implementations; overseeing upgrades and release schedules for BI and database products; reviewing the impact of changes before and after implementation; and communicating project results and updates to senior leadership.
- Provide supervision and development opportunities for associates by hiring and training on modern tools such as Code Puppy and Copilot; mentoring; assigning duties; providing recognition; and ensuring diversity awareness.
- Create and revise complex BI and reporting solutions by identifying business requirements and validating gathered information; collaborating with subject matter experts, data experts, and other BI and analytics experts; coordinating the evaluation of customer and supplier solutions; making recommendations to meet end-user objectives; designing and conducting analysis and reporting solutions; writing detailed dashboard and reporting specifications; documenting and revising data collection processes; and creating reports and visualizations to effectively communicate results.
- Understand current customer BI and reporting needs by managing production of scheduled reports; managing reporting databases to ensure accuracy and completeness of data; maintaining materials for state-of-the-art user interfaces that enable business users to interpret trends and exceptions; providing guidance and direction to stakeholders; and identifying gaps and opportunities in BI and reporting solutions.

### Senior Data Analyst

2022

*Supply Chain, Walmart Inc, Bentonville, USA*

- Developed 20+ dashboards translating complex data patterns into clear visualizations for diverse stakeholders.
- Applied time series forecasting models for transportation demand, balancing complexity with applicability.
- Elevated customer satisfaction from 60% to 95% through rigorous root cause analysis and systematic frameworks.
- Tackled inventory optimization for 1,500+ SKUs using mathematical modeling and constraint optimization.

### Business Analyst Intern

2022

*Aezion Inc, Dallas, USA*

- Applied quantitative risk analysis to \$2B programs using Monte Carlo simulations.
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Achieved 20% supply chain cost reduction through analytical insights.

## Research Analyst

2021

*Southern Methodist University, Dallas, USA*

- Developed mathematical models in Python quantifying marketing impact on consumer behavior.
- Built neural network models predicting performing arts attendance during COVID-19 unprecedented conditions.
- Created an interactive web application (RShiny & SQL) enabling exploration of complex predictive models.

## R&D Quality Assurance Manager

2020 - 2021

*Dassault Systemes Solutions Lab, Pune, India*

- Led a team of 7 automating analytical workflows (Power BI), achieving 10% revenue increase.
- Defined 100+ KPIs for 6 product launches, establishing rigorous quantitative measurement frameworks.
- Created dashboards translating complex clinical trial data into accessible insights for medical stakeholders.

## Senior Consultant

2019 - 2020

*Publicis Sapient, Gurgaon, India*

- Performed EDA using advanced statistical techniques (Q-Q plots, correlation analysis).
- Conducted rigorous root cause analysis improving software quality.

## QA Analyst

2018 - 2019

*Capgemini, Pune, India*

- Led an analytical team developing quantitative models for media supply chain optimization.
- Performed EDA on financial loan data, uncovering behavioral patterns and risk factors.

## Senior Data Analyst

2014 - 2018

*Infosys Ltd, Pune, India*

- Applied hypothesis testing, regression, and correlation for strategic business decisions.
- Performed linear programming and sensitivity analysis for resource optimization.

## ANALYTICAL PROJECTS

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**Phoenix (Quickbase) Finance Platform:** Designed new processes and workflows in Phoenix (Quickbase) to run business and finance operations across all POs, maintaining financial changes in data sourced directly from SAP.

**Code Puppy Diaries (Live):** A living project exploring AI-assisted development with Code Puppy - experiments, automations, and learnings, versioned and shared on Walmart Enterprise GitHub ([gecg01.walmart.com/s0g0em7/Code-Puppy-Diaries-live](https://github.com/gecg01.walmart.com/s0g0em7/Code-Puppy-Diaries-live)).

**EOL Electrical Analysis (Multi-Agent):** Automated end-of-life electrical analysis - picks stores one by one, finds the latest diagram model from a shared folder, downloads it, converts it into a readable Excel form, and pushes the data into Phoenix (Quickbase), all orchestrated by 4 parallel Quickbase agents.

**Power Automate Ticketing Bot:** Built an automation bot in Power Automate that copies comments from an email and pastes them into the DRT associated with a ticket, saving roughly 7 hours every day.

**Statistical Decision Modeling:** Applied hypothesis testing, multiple regression, and correlation for a multi-million dollar acquisition evaluation (Sleep Cool Mattress).

**Consumer Behavior Analysis:** Analyzed large-scale Six Flags data identifying behavioral patterns; built interactive Tableau dashboards.

**Optimization Modeling:** Performed linear programming & sensitivity analysis for LEGO production planning and resource allocation.

## EDUCATION

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### MS in Business Analytics (GPA: 3.7/4.0)

2021 - 2022

*Southern Methodist University, Dallas, USA*

- Coursework: Statistical Analysis, ML, Predictive Modeling, Decision Optimization, Time Series.

### BS in Computer Science & Engineering

2010 - 2014

*Lingaya's University, India*

- Foundation: Algorithms, Data Structures, Computational Math, Database Systems.

## TECHNICAL SKILLS & METHODOLOGIES

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**Programming & Tools:** Python (NumPy, Pandas, Scikit-learn), R (Statistical Analysis), SQL, Tableau, Power BI, GCP BigQuery, Alteryx, Dataiku, Excel (Advanced Analytics), Code Puppy, Git, AI Innovation Lab, Wibey, VS Code

**Analytical Methods:** Hypothesis Testing, Regression (Linear, Logistic, Multiple), Time Series (ARIMA, Prophet), Neural Networks, Decision Trees, Clustering, Monte Carlo Simulation, Linear Programming, Sensitivity Analysis, Root Cause Analysis, A/B Testing, EDA

**Domains:** Supply Chain, Healthcare, Finance, Social Media, Performing Arts, Retail, Facilities Services, Walmart Power

## CERTIFICATIONS & COMMITMENT

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- Certifications: Dataiku Core Designer | Bloomberg Market Concepts | Python Data Science Toolkit (DataCamp) | Data Analytics in R | Lean Six Sigma Yellow Belt (AIGPE)
- Continuous Learning: Rapidly mastered new platforms (Dataiku, Alteryx), demonstrating intellectual curiosity and adaptability.
- Cross-Disciplinary Impact: Applied expertise across supply chain, healthcare, finance, social media, performing arts, and facilities services.
- Collaborative Excellence: Effective communication with technical teams, business stakeholders, and academic researchers.